# D-Flow, QGF and imperfect offline critics

## Morning research report | second campaign | November 6, 2026

Snapshot: 2026-11-06T09:55:50.673531+00:00. Root verified writers finished or safely checkpointed.

This report distinguishes completed 20-episode test cells, completed 8-start validation, analytic points and unfinished work. A pending cell is never a zero. Algorithms were not retuned from test outcomes.

| Study | Validation executions | Test executions | Frozen choices | Test models complete |
|---|---|---|---|---|
| coverage | 11360 | 4560 | 174 | 6 |
| native | 4920 | 0 | 76 | 0 |

At this snapshot there are 174 completed requested primary condition/model test cells, including 22 zero-success cells. Display aliases are marked; these are not independent extra trials.

The completed stress comparison does not show a universal sampler winner. Across trained seeds 11/23/37, raw QGF scored FULL 20/13/20 versus HOLE 20/19/20. Raw MH scored 20/9/19 versus 20/17/20; averaged-Q MH scored 20/11/19 versus 20/16/20. Averaged-Q QGF scored 20/7/20 versus 20/19/20. Each condition used its own validation selection. The HOLE intervention affected Q/V training only; BC retained full behavior data. This data hole did not consistently harm QGF or demonstrate true-Q repair.

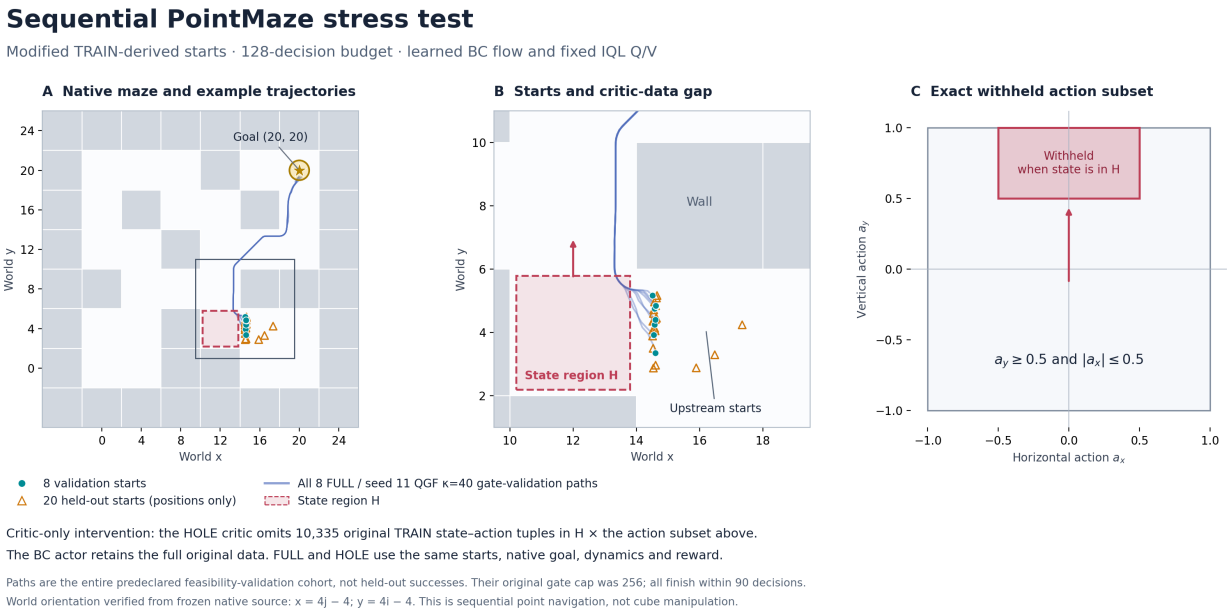
Both regular full-Jacobian/Euler conditions scored 0/20 in all six models. Seed 37 lambda 0 primaries used canonical BC; their frozen positive alternatives were also 0/20. Stochastic Euler lambda 1 scored FULL 0/0/0 versus HOLE 1/0/0; ordinary IPG scored 0/0/0 versus 0/0/5. Standard 1,000-step task 1 stopped at its two-attempt cap with 4,920 completed validation episodes and one partial batch; it has no heldout tests. Task 5 was not started.

HOLE seed 23 QGF entered the joint removed state/action subset in all 20 episodes and succeeded in 19. Its success is therefore not explained solely by avoiding that subset. This observation does not establish that its Q predictions are calibrated.

Main design: 12 D-Flow families produce 22 fixed-lambda conditions, plus 3 QGF and 4 particle conditions. Each condition selects its own settings on validation. Learned-Q search and simulator-validation budgets are unequal; selections are search-limited, not global optima.

**The actor is an x,y-conditioned BC flow, without explicit goal input. IQL-trained Q/V are reused with that separately frozen BC actor, not an IQL advantage-weighted actor. BC 0/8 alone does not establish poor fit to the behavior distribution or diagnose a goal-conditioned actor.**

# The task: a point robot navigating the maze




Native MuJoCo PointMaze has real next states and rewards. The modified primary stress task places the point robot at predeclared upstream TRAIN-derived positions, aims at goal (20,20), and limits each episode to 128 decisions. The figure shows validation/test start positions only and all 8 feasibility-validation paths from FULL seed 11 using fixed QGF kappa=40; no heldout success is depicted.

The HOLE critic excludes 10335 complete training tuples in max(|x-12|,|y-4|)<=1.8 AND ay>=.5 AND |ax|<=.5. This is a rectangular upward-action subset. FULL keeps those tuples; both use the full-data BC actor. Missing critic coverage is an intervention, not proof of Q error.

The FULL-QGF-only feasibility gate selected the largest qualifying budget, 128: 20/24 goal successes and 24/24 episodes entering the joint state/action hole. No HOLE, new sampler or test outcome chose this budget. This modified reset/budget task is separate from the original 1000-step benchmark tasks 1/5.

Training episodes for anchor/prescreen/validation/test mapping are disjoint by declaration. Exact starts and policy roots are in the saved DESIGN cohorts/reset map. Native episodes are separate from analytic final-point trials and prior MuJoCo cube manipulation; no new robot grasp sweep was launched on weak actors.

# The 12 D-Flow families and additional baselines

| # | Family | Source noise | Endpoint Jacobian | Guidance endpoint | Score |
|---|---|---|---|---|---|
| 1 | DF R full/full | initial only | full | full | raw Q |
| 2 | DF R full/euler | initial only | full | euler | raw Q |
| 3 | DF R identity/full | initial only | identity | full | raw Q |
| 4 | DF R identity/euler | initial only | identity | euler | raw Q |
| 5 | DF S full/full | yes | full | full | raw Q |
| 6 | DF S full/euler | yes | full | euler | raw Q |
| 7 | DF S identity/full | yes | identity | full | raw Q |
| 8 | DF S identity/euler | yes | identity | euler | raw Q |
| 9 | DF avgQ R full/full | initial only | full | full | avg-Q |
| 10 | DF avgQ S full/full | yes | full | full | avg-Q |
| 11 | DF MH full/full | yes | full | full | raw Q + MH |
| 12 | DF avgQ MH full/full | yes | full | full | avg-Q + MH |

Label legend: R means regular source updates; S means stochastic updates with fresh movement noise; L0/L1 mean lambda=0/1. full/identity describe the endpoint Jacobian, and full/Euler describe the endpoint used for guidance. avgQ means action-space Q averaging; MH means an accept/reject correction.

The first 10 families each have lambda 0/1 as separately fixed strata; raw and averaged MH fix lambda=temperature=1. The original eight cross regular/stochastic, full/identity endpoint Jacobians, full-flow/Euler guidance. All execute the full BC map after source updates.

Additional conditions: ordinary QGF, action-space averaged-Q QGF, separate Jacobian-smoothing QGF, IPG, IPG-CV, residual-weighted IPG and FK. Controls include canonical BC, uniform BC particle populations for N=8/16 and M=8/32, QGF fixed at .02/.04, greedy 81-action Q, and fixed 40 QGF in coverage.

Particle methods generate a small group of candidate actions at each state. IPG moves the group using a kernel-based Q correction; IPG-CV uses a centered version of that kernel system. IPG-RW also keeps residual importance weights and can resample when they concentrate. FK reweights and resamples without the IPG transport correction. IPG/CV choose a particle uniformly; IPG-RW/FK sample from their final weights. They do not simply choose the highest-Q candidate.

Source diffusion, nearby-action Q averaging, paper Jacobian smoothing and MH are distinct operations. All sampler cells retain the same critic/BC weights within each task/arm/trained model. FULL and HOLE intentionally use different critics. Labels identify the requested condition; its actual executed configuration can be canonical BC.

# Coefficients, proxy targets and executed actions

**Let C=clip[-1,1], F(z)=C[complete M-step BC integration], E(z)=C[z+v(s,z,0)]. Every D-Flow variant executes F(zI). At fixed physical state, source updates are:**

z_next = z + eta [kappa g(z) - lambda z] + sqrt(2 eta Ttemp) epsilon

The noise term is omitted in regular variants. Lambda pulls toward source origin 0, not toward the particular initial draw or a hard data-support set. I is outer source iterations; M is flow pseudo-time slices; neither is a physical step or episode. Primary stochastic Ttemp=1.

| Variant | Guidance field g | Optimized score / execution |
|---|---|---|
| full-J/full | J_F transpose grad Q(F) | Q(F); execute F |
| full-J/Euler | J_E transpose grad Q(E) | Q(E); execute F |
| identity/full | action grad Q(F) | surrogate source field; execute F |
| identity/Euler | action grad Q(E) | surrogate source field; execute F |

Full derivatives include the selected endpoint clip; identity bypasses it. Identity fields need not be conservative. The full-J/Euler continuous target involves Q(E), so its executed F-pushforward is generally not BC tilted by Q at the executed action.

For deterministic lambda 0, equal eta*kappa at fixed I/map/source/bank gives identical mathematical iterates. This does not guarantee float32 bitwise identity. Active prior/noise breaks a trade of only eta*kappa; equal eta*kappa*I at different I is not exact for nonlinear guidance.

Qbar(a)=mean_k min_head Q(s,C[a+sigma epsilon_k]). The antithetic K=4/16 panel is fixed for the whole action call and both MH sides. It is a finite-panel surrogate. Sigma changes the action-space score, not source temperature; sigma=0 bypasses the additional perturbation clip to preserve raw derivatives.

QGF refreshes v, a_hat=C[x+(1-t)v] and grad Q(a_hat) at every pseudo-time slice, then x_next=x+(v+kappa grad Q)/M. Standard QGF uses identity endpoint Jacobian and no source eta/lambda/I loop. Numeric kappa has no cross-method or cross-critic trust meaning.

MH adds the exact ideal-arithmetic forward/reverse Gaussian ratio for log pi=kappa Q(F)-||z||^2/2. It corrects transitions for that specified target, not Q. Finite I need not mix. Unadjusted finite-step Langevin is not exact posterior sampling; stochastic lambda 0 is noisy finite-horizon search, not a normalized prior-free source posterior. Full appendix: review/coefficient_semantics/APPENDIX.txt.

In plain language, MH proposes a noisy source move, then accepts it or keeps the current source. After either decision, the next iteration continues; rejection does not end action generation.

# Selection, physical counts and unfinished work

| Study/model | Frozen | Validation | Test | Status |
|---|---|---|---|---|
| coverage full s11 | 29/29 | 1904 | 760 | test complete |
| coverage full s23 | 29/29 | 1888 | 760 | test complete |
| coverage full s37 | 29/29 | 1888 | 760 | test complete |
| coverage hole s11 | 29/29 | 1896 | 760 | test complete |
| coverage hole s23 | 29/29 | 1896 | 760 | test complete |
| coverage hole s37 | 29/29 | 1888 | 760 | test complete |
| native 1 s11 | 29/29 | 1896 | 0 | validation complete |
| native 1 s23 | 29/29 | 1896 | 0 | validation complete |
| native 1 s37 | 18/29 | 1128 | 0 | started/partial |
| native 5 s11 | 0/29 | 0 | 0 | not started |
| native 5 s23 | 0/29 | 0 | 0 | not started |
| native 5 s37 | 0/29 | 0 | 0 | not started |

Declared domains: kappa 0, .02,.04,.06,.08,.1,.2,.3,.4,.6,.8,1,2,3,5,8,12,20,40,80; eta .005/.01/.03/.1; I 1/2/4/8/16/32/64/128/256; sigma .01/.03/.1/.2; K 4/16. Raw DF screens 720 settings; averaged DF uses reference Stage A plus predeclared bounded Stage B, not every interaction. Raw physical shortlists<=7, averaged<=15; QGF validates all 20 kappas.

Each task/arm/trained seed/family/fixed lambda owns its selection. Canonical BC is eligible. A positive secondary is frozen when a primary has no positive guidance. All 174 coverage choices freeze before any arm test; native freezes all 3 models separately for each task. 20 fresh paired starts test each requested unique exact config. Simulator validation is an oracle tuning resource.

Actual executions are counts of completed saved batches only. Partial checkpoints remain resumable and excluded. Display aliases reuse identical config IDs; numeric int/float aliases are tracked separately. Correlated trajectories, paired starts and replay are not extra training seeds. The latest passing audit has matching branch design identities and model counts/phase/choice/display summaries for this snapshot: True. This summary comparison is not an independent all-array bytewise audit.

Completed episodes with nonfinite saved returns: 0. Failed outcomes are retained, not discarded or replaced by zero. Matching NaN CSV/metadata returns are accepted explicitly; the exact JSON snapshot encodes nonfinite floats with a named tag and CSV supplements retain nan/inf plus counts. Live progress is captured separately and does not establish immutable results.

# Primary coverage tests: FULL and HOLE, by trained seed

| Requested condition | FULL s11 | FULL s23 | FULL s37 | HOLE s11 | HOLE s23 | HOLE s37 |
|---|---|---|---|---|---|---|
| QGF | 20/20 | 13/20 | 20/20 | 20/20 | 19/20 | 20/20 |
| DF R full/full L0 | 20/20 | 6/20 | 20/20 | 20/20 | 15/20 | 20/20 |
| DF R full/full L1 | 20/20 | 12/20 | 20/20 | 20/20 | 17/20 | 20/20 |
| DF R full/euler L0 | 0/20 | 0/20 | 0/20* | 0/20 | 0/20 | 0/20* |
| DF R full/euler L1 | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 |
| DF R identity/full L0 | 20/20 | 19/20 | 20/20 | 20/20 | 19/20 | 20/20 |
| DF R identity/full L1 | 20/20 | 19/20 | 20/20 | 20/20 | 16/20 | 20/20 |
| DF R identity/euler L0 | 20/20 | 19/20 | 20/20 | 20/20 | 16/20 | 20/20 |
| DF R identity/euler L1 | 20/20 | 18/20 | 20/20 | 20/20 | 17/20 | 20/20 |
| DF S full/full L0 | 20/20 | 12/20 | 20/20 | 20/20 | 16/20 | 20/20 |
| DF S full/full L1 | 20/20 | 11/20 | 20/20 | 20/20 | 16/20 | 20/20 |
| DF S full/euler L0 | 15/20 | 14/20 | 11/20 | 17/20 | 14/20 | 18/20 |
| DF S full/euler L1 | 0/20 | 0/20 | 0/20 | 1/20 | 0/20 | 0/20 |
| DF S identity/full L0 | 20/20 | 12/20 | 20/20 | 20/20 | 15/20 | 20/20 |
| DF S identity/full L1 | 20/20 | 16/20 | 20/20 | 20/20 | 18/20 | 20/20 |
| DF S identity/euler L0 | 20/20 | 12/20 | 20/20 | 20/20 | 12/20 | 20/20 |
| DF S identity/euler L1 | 20/20 | 14/20 | 20/20 | 20/20 | 19/20 | 20/20 |
| DF avgQ R full/full L0 | 20/20 | 6/20 | 19/20 | 20/20 | 15/20 | 20/20 |
| DF avgQ R full/full L1 | 20/20 | 10/20 | 20/20 | 20/20 | 17/20 | 20/20 |
| DF avgQ S full/full L0 | 20/20 | 12/20 | 20/20 | 20/20 | 15/20 | 20/20 |
| DF avgQ S full/full L1 | 20/20 | 11/20 | 20/20 | 20/20 | 16/20 | 20/20 |
| DF MH full/full | 20/20 | 9/20 | 19/20 | 20/20 | 17/20 | 20/20 |
| DF avgQ MH full/full | 20/20 | 11/20 | 19/20 | 20/20 | 16/20 | 20/20 |
| QGF avg-Q | 20/20 | 7/20 | 20/20 | 20/20 | 19/20 | 20/20 |
| QGF smooth-J | 20/20 | 10/20 | 20/20 | 20/20 | 14/20 | 20/20 |
| IPG | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | 5/20 |
| IPG-CV | 17/20 | 4/20 | 8/20 | 19/20 | 1/20 | 14/20 |
| IPG-RW | 17/20 | 1/20 | 3/20 | 19/20 | 1/20 | 10/20 |
| FK | 15/20 | 11/20 | 11/20 | 17/20 | 7/20 | 16/20 |

Every completed value is successes/20 from the predeclared test cohort. pending means not completed or not selected; partial means an unfinished saved test batch. A star marks an exact config-ID display alias, not a separate trial. All zeros shown are completed zeros. No mean across missing trained seeds is fabricated.

# Controls, positive secondaries and alias bookkeeping

| Requested control | FULL s11 | FULL s23 | FULL s37 | HOLE s11 | HOLE s23 | HOLE s37 |
|---|---|---|---|---|---|---|
| Canonical BC | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 |
| Uniform BC N8_T8 | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 |
| Uniform BC N8_T32 | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 |
| Uniform BC N16_T8 | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 |
| Uniform BC N16_T32 | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 |
| QGF fixed0.02 | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 |
| QGF fixed0.04 | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 |
| Greedy81 Q | 20/20 | 0/20 | 20/20 | 20/20 | 17/20 | 20/20 |
| QGF fixed40 | 20/20 | 8/20 | 20/20 | 20/20 | 19/20 | 20/20 |

| Arm/seed | Requested secondary | Own settings | Outcome |
|---|---|---|---|
| full s37 | DF R full/euler L0 | k0.08 e0.1 I8 L0 | 0/20 |
| hole s37 | DF R full/euler L0 | k0.08 e0.1 I8 L0 | 0/20 |

coverage: 230 completed displayed requested rows, 2 exact-ID display aliases, 2 completed positive secondaries. These row counts are separate from actual physical batch/episode executions.

native: 0 completed displayed requested rows, 0 exact-ID display aliases, 0 completed positive secondaries. These row counts are separate from actual physical batch/episode executions.

Canonical BC means QGF kappa 0 with no outer source evolution. Finite-I D-Flow kappa 0 can still move under prior/noise; it is not silently called BC. At ideal-arithmetic MH kappa 0, marginal BC preservation does not mean the same paired action. Numeric policy equality requires more evidence than matching display labels.

Full requested rows, actual executed families, config IDs, alias links, test statuses and every active/unused parameter field are in the CSV supplements. Numeric groups and repeated trajectory records remain distinct from verified array aliases; consult the bound independent saved-artifact audit.

# Own validation-frozen settings: coverage trained seed 11

Each arm selected independently. k=kappa; e=eta; I=source iterations; L=lambda; s=sigma; K=average samples; N=particles; M=flow slices; n=particle noise; r=ridge. Unused placeholders are omitted. Canonical BC is stated explicitly; its requested family label is retained.

| Requested condition | FULL own settings | HOLE own settings |
|---|---|---|
| QGF | k40 | k40 |
| DF R full/full L0 | k80 e0.1 I256 L0 | k80 e0.1 I128 L0 |
| DF R full/full L1 | k80 e0.03 I256 L1 | k80 e0.1 I32 L1 |
| DF R full/euler L0 | k0.1 e0.1 I256 L0 | k0.1 e0.1 I256 L0 |
| DF R full/euler L1 | k80 e0.01 I2 L1 | k80 e0.01 I2 L1 |
| DF R identity/full L0 | k5 e0.1 I256 L0 | k5 e0.1 I256 L0 |
| DF R identity/full L1 | k80 e0.1 I64 L1 | k80 e0.1 I256 L1 |
| DF R identity/euler L0 | k5 e0.1 I256 L0 | k80 e0.1 I64 L0 |
| DF R identity/euler L1 | k80 e0.03 I256 L1 | k80 e0.1 I128 L1 |
| DF S full/full L0 | k80 e0.1 I128 L0 | k80 e0.1 I256 L0 |
| DF S full/full L1 | k80 e0.01 I256 L1 | k80 e0.01 I256 L1 |
| DF S full/euler L0 | k5 e0.1 I256 L0 | k5 e0.1 I256 L0 |
| DF S full/euler L1 | k5 e0.1 I16 L1 | k80 e0.1 I64 L1 |
| DF S identity/full L0 | k80 e0.1 I128 L0 | k80 e0.1 I64 L0 |
| DF S identity/full L1 | k80 e0.1 I128 L1 | k80 e0.03 I128 L1 |
| DF S identity/euler L0 | k40 e0.1 I128 L0 | k80 e0.1 I16 L0 |
| DF S identity/euler L1 | k80 e0.1 I128 L1 | k80 e0.03 I128 L1 |
| DF avgQ R full/full L0 | k80 e0.1 I256 L0 s0.03 K4 | k80 e0.1 I256 L0 s0.03 K16 |
| DF avgQ R full/full L1 | k80 e0.03 I256 L1 s0.01 K4 | k80 e0.03 I128 L1 s0.01 K4 |
| DF avgQ S full/full L0 | k80 e0.1 I256 L0 s0.03 K4 | k80 e0.1 I256 L0 s0.01 K16 |
| DF avgQ S full/full L1 | k80 e0.01 I256 L1 s0.03 K4 | k80 e0.01 I256 L1 s0.03 K16 |
| DF MH full/full | k80 e0.1 I128 L1 | k80 e0.1 I128 L1 |
| DF avgQ MH full/full | k80 e0.03 I128 L1 s0.03 K4 | k80 e0.1 I128 L1 s0.03 K16 |
| QGF avg-Q | k80 s0.2 K4 | k80 s0.2 K16 |
| QGF smooth-J | k80 s0.2 K4 | k80 s0.2 K16 |
| IPG | k5 N16 M8 n0.005 r0.01 | k5 N16 M8 n0.02 r0.01 |
| IPG-CV | k40 N16 M8 n0.005 r0.01 | k40 N16 M8 n0.02 r0.01 |
| IPG-RW | k40 N16 M32 n0 r0.1 | k40 N16 M32 n0 r0.1 |
| FK | k80 N16 M32 n0.02 | k80 N16 M32 n0.02 |

A setting at k80 or I256 is at a declared grid boundary, not a proven optimum. All exact numbers, original JSON numeric types and primary/positive roles are retained in MORNING_SELECTED_SETTINGS.csv and the exact snapshot JSON.

# Own validation-frozen settings: coverage trained seed 23

Each arm selected independently. k=kappa; e=eta; I=source iterations; L=lambda; s=sigma; K=average samples; N=particles; M=flow slices; n=particle noise; r=ridge. Unused placeholders are omitted. Canonical BC is stated explicitly; its requested family label is retained.

| Requested condition | FULL own settings | HOLE own settings |
|---|---|---|
| QGF | k20 | k40 |
| DF R full/full L0 | k1 e0.1 I256 L0 | k80 e0.1 I256 L0 |
| DF R full/full L1 | k80 e0.005 I256 L1 | k80 e0.01 I256 L1 |
| DF R full/euler L0 | k40 e0.1 I256 L0 | k40 e0.1 I256 L0 |
| DF R full/euler L1 | k80 e0.1 I64 L1 | k80 e0.1 I64 L1 |
| DF R identity/full L0 | k1 e0.1 I256 L0 | k5 e0.1 I256 L0 |
| DF R identity/full L1 | k80 e0.1 I256 L1 | k8 e0.1 I64 L1 |
| DF R identity/euler L0 | k1 e0.1 I256 L0 | k0.4 e0.1 I256 L0 |
| DF R identity/euler L1 | k80 e0.03 I256 L1 | k12 e0.005 I256 L1 |
| DF S full/full L0 | k5 e0.1 I256 L0 | k80 e0.03 I16 L0 |
| DF S full/full L1 | k80 e0.005 I256 L1 | k80 e0.005 I256 L1 |
| DF S full/euler L0 | k5 e0.1 I256 L0 | k5 e0.1 I256 L0 |
| DF S full/euler L1 | k80 e0.1 I32 L1 | k80 e0.1 I32 L1 |
| DF S identity/full L0 | k5 e0.03 I256 L0 | k5 e0.03 I256 L0 |
| DF S identity/full L1 | k80 e0.1 I256 L1 | k80 e0.1 I256 L1 |
| DF S identity/euler L0 | k5 e0.03 I256 L0 | k3 e0.1 I256 L0 |
| DF S identity/euler L1 | k80 e0.1 I256 L1 | k80 e0.1 I256 L1 |
| DF avgQ R full/full L0 | k1 e0.1 I256 L0 s0.01 K4 | k80 e0.1 I256 L0 s0.03 K4 |
| DF avgQ R full/full L1 | k80 e0.005 I256 L1 s0.1 K4 | k80 e0.005 I256 L1 s0.2 K4 |
| DF avgQ S full/full L0 | k5 e0.1 I256 L0 s0.01 K4 | k40 e0.1 I16 L0 s0.2 K4 |
| DF avgQ S full/full L1 | k80 e0.005 I256 L1 s0.1 K16 | k80 e0.005 I256 L1 s0.1 K16 |
| DF MH full/full | k80 e0.1 I256 L1 | k80 e0.03 I256 L1 |
| DF avgQ MH full/full | k80 e0.005 I256 L1 s0.2 K4 | k80 e0.005 I256 L1 s0.2 K4 |
| QGF avg-Q | k5 s0.01 K4 | k80 s0.2 K16 |
| QGF smooth-J | k12 s0.2 K4 | k12 s0.2 K4 |
| IPG | k5 N16 M8 n0.02 r0.01 | k5 N16 M8 n0.02 r0.01 |
| IPG-CV | k40 N16 M8 n0.02 r0.01 | k40 N16 M8 n0.005 r0.01 |
| IPG-RW | k40 N16 M8 n0.005 r0.1 | k40 N16 M8 n0.02 r0.1 |
| FK | k80 N16 M32 n0.02 | k80 N16 M32 n0.02 |

A setting at k80 or I256 is at a declared grid boundary, not a proven optimum. All exact numbers, original JSON numeric types and primary/positive roles are retained in MORNING_SELECTED_SETTINGS.csv and the exact snapshot JSON.

# Own validation-frozen settings: coverage trained seed 37

Each arm selected independently. k=kappa; e=eta; I=source iterations; L=lambda; s=sigma; K=average samples; N=particles; M=flow slices; n=particle noise; r=ridge. Unused placeholders are omitted. Canonical BC is stated explicitly; its requested family label is retained.

| Requested condition | FULL own settings | HOLE own settings |
|---|---|---|
| QGF | k40 | k40 |
| DF R full/full L0 | k20 e0.1 I256 L0 | k20 e0.1 I256 L0 |
| DF R full/full L1 | k80 e0.03 I256 L1 | k80 e0.03 I128 L1 |
| DF R full/euler L0 | Canonical BC (k0) | Canonical BC (k0) |
| DF R full/euler L1 | k80 e0.1 I64 L1 | k80 e0.1 I128 L1 |
| DF R identity/full L0 | k5 e0.1 I256 L0 | k40 e0.03 I256 L0 |
| DF R identity/full L1 | k80 e0.03 I256 L1 | k80 e0.03 I128 L1 |
| DF R identity/euler L0 | k5 e0.1 I256 L0 | k80 e0.03 I128 L0 |
| DF R identity/euler L1 | k80 e0.03 I256 L1 | k80 e0.01 I256 L1 |
| DF S full/full L0 | k80 e0.1 I256 L0 | k80 e0.1 I256 L0 |
| DF S full/full L1 | k80 e0.01 I256 L1 | k80 e0.01 I256 L1 |
| DF S full/euler L0 | k5 e0.1 I256 L0 | k12 e0.1 I256 L0 |
| DF S full/euler L1 | k40 e0.1 I256 L1 | k40 e0.03 I256 L1 |
| DF S identity/full L0 | k80 e0.03 I128 L0 | k80 e0.1 I128 L0 |
| DF S identity/full L1 | k80 e0.03 I128 L1 | k80 e0.03 I128 L1 |
| DF S identity/euler L0 | k80 e0.03 I128 L0 | k80 e0.1 I128 L0 |
| DF S identity/euler L1 | k80 e0.03 I128 L1 | k80 e0.03 I128 L1 |
| DF avgQ R full/full L0 | k40 e0.1 I256 L0 s0.03 K16 | k80 e0.03 I256 L0 s0.2 K16 |
| DF avgQ R full/full L1 | k80 e0.03 I128 L1 s0.01 K4 | k80 e0.03 I256 L1 s0.01 K4 |
| DF avgQ S full/full L0 | k80 e0.1 I256 L0 s0.2 K4 | k80 e0.1 I256 L0 s0.2 K4 |
| DF avgQ S full/full L1 | k80 e0.01 I256 L1 s0.2 K16 | k80 e0.01 I256 L1 s0.2 K4 |
| DF MH full/full | k80 e0.01 I256 L1 | k80 e0.01 I256 L1 |
| DF avgQ MH full/full | k80 e0.01 I256 L1 s0.03 K16 | k80 e0.1 I128 L1 s0.03 K16 |
| QGF avg-Q | k80 s0.1 K16 | k80 s0.2 K4 |
| QGF smooth-J | k40 s0.2 K16 | k40 s0.2 K16 |
| IPG | k40 N16 M8 n0.02 r0.01 | k40 N16 M8 n0.02 r0.01 |
| IPG-CV | k40 N16 M8 n0.02 r0.01 | k40 N16 M8 n0 r0.01 |
| IPG-RW | k40 N16 M8 n0.02 r0.1 | k40 N16 M8 n0.02 r0.1 |
| FK | k80 N16 M32 n0.02 | k80 N16 M32 n0.02 |

A setting at k80 or I256 is at a declared grid boundary, not a proven optimum. All exact numbers, original JSON numeric types and primary/positive roles are retained in MORNING_SELECTED_SETTINGS.csv and the exact snapshot JSON.

# Standard native tasks 1/5: completed and partial cells

| Requested condition | t1 s11 | t1 s23 | t1 s37 | t5 s11 | t5 s23 | t5 s37 |
|---|---|---|---|---|---|---|
| QGF | pending | pending | pending | pending | pending | pending |
| DF R full/full L0 | pending | pending | pending | pending | pending | pending |
| DF R full/full L1 | pending | pending | pending | pending | pending | pending |
| DF R full/euler L0 | pending | pending | pending | pending | pending | pending |
| DF R full/euler L1 | pending | pending | pending | pending | pending | pending |
| DF R identity/full L0 | pending | pending | pending | pending | pending | pending |
| DF R identity/full L1 | pending | pending | pending | pending | pending | pending |
| DF R identity/euler L0 | pending | pending | pending | pending | pending | pending |
| DF R identity/euler L1 | pending | pending | pending | pending | pending | pending |
| DF S full/full L0 | pending | pending | pending | pending | pending | pending |
| DF S full/full L1 | pending | pending | pending | pending | pending | pending |
| DF S full/euler L0 | pending | pending | pending | pending | pending | pending |
| DF S full/euler L1 | pending | pending | pending | pending | pending | pending |
| DF S identity/full L0 | pending | pending | pending | pending | pending | pending |
| DF S identity/full L1 | pending | pending | pending | pending | pending | pending |
| DF S identity/euler L0 | pending | pending | pending | pending | pending | pending |
| DF S identity/euler L1 | pending | pending | pending | pending | pending | pending |
| DF avgQ R full/full L0 | pending | pending | pending | pending | pending | pending |
| DF avgQ R full/full L1 | pending | pending | pending | pending | pending | pending |
| DF avgQ S full/full L0 | pending | pending | pending | pending | pending | pending |
| DF avgQ S full/full L1 | pending | pending | pending | pending | pending | pending |
| DF MH full/full | pending | pending | pending | pending | pending | pending |
| DF avgQ MH full/full | pending | pending | pending | pending | pending | pending |
| QGF avg-Q | pending | pending | pending | pending | pending | pending |
| QGF smooth-J | pending | pending | pending | pending | pending | pending |
| IPG | pending | pending | pending | pending | pending | pending |
| IPG-CV | pending | pending | pending | pending | pending | pending |
| IPG-RW | pending | pending | pending | pending | pending | pending |
| FK | pending | pending | pending | pending | pending | pending |

These retain their native 1000-step task/reset semantics and are distinct from the modified 128-step stress. The full standard grid may be partial because its measured cost exceeds one night. Partial grids are not algorithm rankings. All native controls and positive secondaries are retained in MORNING_TEST_RESULTS.csv.

# Own frozen native settings: task 1

| Requested condition | Trained seed 11 | Trained seed 23 | Trained seed 37 |
|---|---|---|---|
| QGF | k80 | k80 | k40 |
| DF R full/full L0 | k5 e0.1 I256 L0 | k20 e0.1 I256 L0 | k5 e0.1 I256 L0 |
| DF R full/full L1 | k5 e0.03 I256 L1 | k80 e0.01 I128 L1 | k80 e0.03 I256 L1 |
| DF R full/euler L0 | k1 e0.1 I256 L0 | Canonical BC (k0) | Canonical BC (k0) |
| DF R full/euler L1 | k5 e0.1 I4 L1 | k0 e0.01 I32 L1 | Canonical BC (k0) |
| DF R identity/full L0 | k5 e0.1 I256 L0 | k40 e0.1 I256 L0 | k5 e0.1 I256 L0 |
| DF R identity/full L1 | k5 e0.1 I128 L1 | k80 e0.03 I256 L1 | k80 e0.03 I256 L1 |
| DF R identity/euler L0 | k20 e0.1 I256 L0 | k20 e0.1 I256 L0 | k5 e0.1 I256 L0 |
| DF R identity/euler L1 | k80 e0.03 I128 L1 | k80 e0.03 I256 L1 | k80 e0.03 I256 L1 |
| DF S full/full L0 | k80 e0.005 I256 L0 | k40 e0.1 I128 L0 | k80 e0.03 I128 L0 |
| DF S full/full L1 | k80 e0.005 I256 L1 | k80 e0.01 I128 L1 | k80 e0.01 I128 L1 |
| DF S full/euler L0 | k5 e0.1 I256 L0 | k5 e0.1 I256 L0 | k12 e0.1 I256 L0 |
| DF S full/euler L1 | k80 e0.1 I128 L1 | k80 e0.1 I256 L1 | k40 e0.1 I128 L1 |
| DF S identity/full L0 | k80 e0.01 I128 L0 | k80 e0.03 I256 L0 | k80 e0.1 I256 L0 |
| DF S identity/full L1 | k80 e0.01 I128 L1 | k80 e0.03 I128 L1 | k80 e0.03 I128 L1 |
| DF S identity/euler L0 | k80 e0.01 I128 L0 | k80 e0.03 I256 L0 | k80 e0.03 I128 L0 |
| DF S identity/euler L1 | k80 e0.01 I128 L1 | k80 e0.03 I128 L1 | k80 e0.03 I128 L1 |
| DF avgQ R full/full L0 | k40 e0.1 I256 L0 s0.2 K4 | k80 e0.03 I256 L0 s0.2 K4 | k20 e0.1 I256 L0 s0.2 K4 |
| DF avgQ R full/full L1 | k80 e0.03 I64 L1 s0.03 K4 | k80 e0.01 I256 L1 s0.01 K16 | not frozen |
| DF avgQ S full/full L0 | k80 e0.03 I256 L0 s0.2 K4 | k80 e0.03 I128 L0 s0.1 K16 | not frozen |
| DF avgQ S full/full L1 | k80 e0.03 I128 L1 s0.01 K16 | k80 e0.01 I128 L1 s0.01 K16 | not frozen |
| DF MH full/full | k80 e0.1 I256 L1 | k80 e0.1 I32 L1 | not frozen |
| DF avgQ MH full/full | k80 e0.1 I256 L1 s0.01 K16 | k80 e0.1 I32 L1 s0.1 K16 | not frozen |
| QGF avg-Q | k80 s0.1 K16 | k80 s0.01 K4 | not frozen |
| QGF smooth-J | k40 s0.1 K4 | k40 s0.2 K16 | not frozen |
| IPG | k80 N8 M32 n0.005 r0.1 | k40 N16 M32 n0.02 r0.01 | not frozen |
| IPG-CV | k20 N16 M8 n0.02 r0.01 | k40 N16 M8 n0.02 r0.01 | not frozen |
| IPG-RW | k5 N16 M8 n0.005 r0.01 | k40 N16 M32 n0.02 r0.1 | not frozen |
| FK | k80 N16 M32 n0.02 | k80 N16 M32 n0.02 | not frozen |

not frozen means selection incomplete or task not started, not a failure score. These settings are chosen on validation only. Per-model test outcomes and all primary/positive/control configurations are separate CSV records.

# Own frozen native settings: task 5

| Requested condition | Trained seed 11 | Trained seed 23 | Trained seed 37 |
|---|---|---|---|
| QGF | not frozen | not frozen | not frozen |
| DF R full/full L0 | not frozen | not frozen | not frozen |
| DF R full/full L1 | not frozen | not frozen | not frozen |
| DF R full/euler L0 | not frozen | not frozen | not frozen |
| DF R full/euler L1 | not frozen | not frozen | not frozen |
| DF R identity/full L0 | not frozen | not frozen | not frozen |
| DF R identity/full L1 | not frozen | not frozen | not frozen |
| DF R identity/euler L0 | not frozen | not frozen | not frozen |
| DF R identity/euler L1 | not frozen | not frozen | not frozen |
| DF S full/full L0 | not frozen | not frozen | not frozen |
| DF S full/full L1 | not frozen | not frozen | not frozen |
| DF S full/euler L0 | not frozen | not frozen | not frozen |
| DF S full/euler L1 | not frozen | not frozen | not frozen |
| DF S identity/full L0 | not frozen | not frozen | not frozen |
| DF S identity/full L1 | not frozen | not frozen | not frozen |
| DF S identity/euler L0 | not frozen | not frozen | not frozen |
| DF S identity/euler L1 | not frozen | not frozen | not frozen |
| DF avgQ R full/full L0 | not frozen | not frozen | not frozen |
| DF avgQ R full/full L1 | not frozen | not frozen | not frozen |
| DF avgQ S full/full L0 | not frozen | not frozen | not frozen |
| DF avgQ S full/full L1 | not frozen | not frozen | not frozen |
| DF MH full/full | not frozen | not frozen | not frozen |
| DF avgQ MH full/full | not frozen | not frozen | not frozen |
| QGF avg-Q | not frozen | not frozen | not frozen |
| QGF smooth-J | not frozen | not frozen | not frozen |
| IPG | not frozen | not frozen | not frozen |
| IPG-CV | not frozen | not frozen | not frozen |
| IPG-RW | not frozen | not frozen | not frozen |
| FK | not frozen | not frozen | not frozen |

not frozen means selection incomplete or task not started, not a failure score. These settings are chosen on validation only. Per-model test outcomes and all primary/positive/control configurations are separate CSV records.

# Validation-only diagnosis: what repeated across FULL models?

Independent saved-artifact reports preserve all 29 selected results and exact settings for FULL seeds 11/23/37. Respectively 24/29, 6/29 and 21/29 conditions reached 8/8. These are tuning outcomes on shared starts, not heldout algorithm wins.

Seed 11: 29/29 FULL/HOLE primary validation success counts match. HOLE QGF=8/8 at its own kappa 40. Own settings can differ; these shared tuning starts can be ceiling-limited. The critic data hole guarantees neither Q error nor QGF failure.

Seed 23: 14/29 FULL/HOLE primary validation success counts match. HOLE QGF=8/8 at its own kappa 40. Own settings can differ; these shared tuning starts can be ceiling-limited. The critic data hole guarantees neither Q error nor QGF failure.

Seed 37: 25/29 FULL/HOLE primary validation success counts match. HOLE QGF=8/8 at its own kappa 40. Own settings can differ; these shared tuning starts can be ceiling-limited. The critic data hole guarantees neither Q error nor QGF failure.

| Selected condition | s11 /8 | s23 /8 | s37 /8 |
|---|---|---|---|
| Regular full-J/Euler L0 | 0 | 0 | 0 |
| Regular full-J/Euler L1 | 0 | 0 | 0 |
| Stochastic full-J/Euler L1 | 0 | 0 | 0 |
| Stochastic full-J/Euler L0 | 7 | 6 | 6 |
| IPG | 0 | 0 | 1 |
| IPG-CV | 8 | 1 | 5 |
| IPG-RW | 8 | 0 | 4 |
| FK | 8 | 2 | 4 |
| Raw MH | 8 | 4 | 8 |
| Averaged MH | 8 | 7 | 8 |

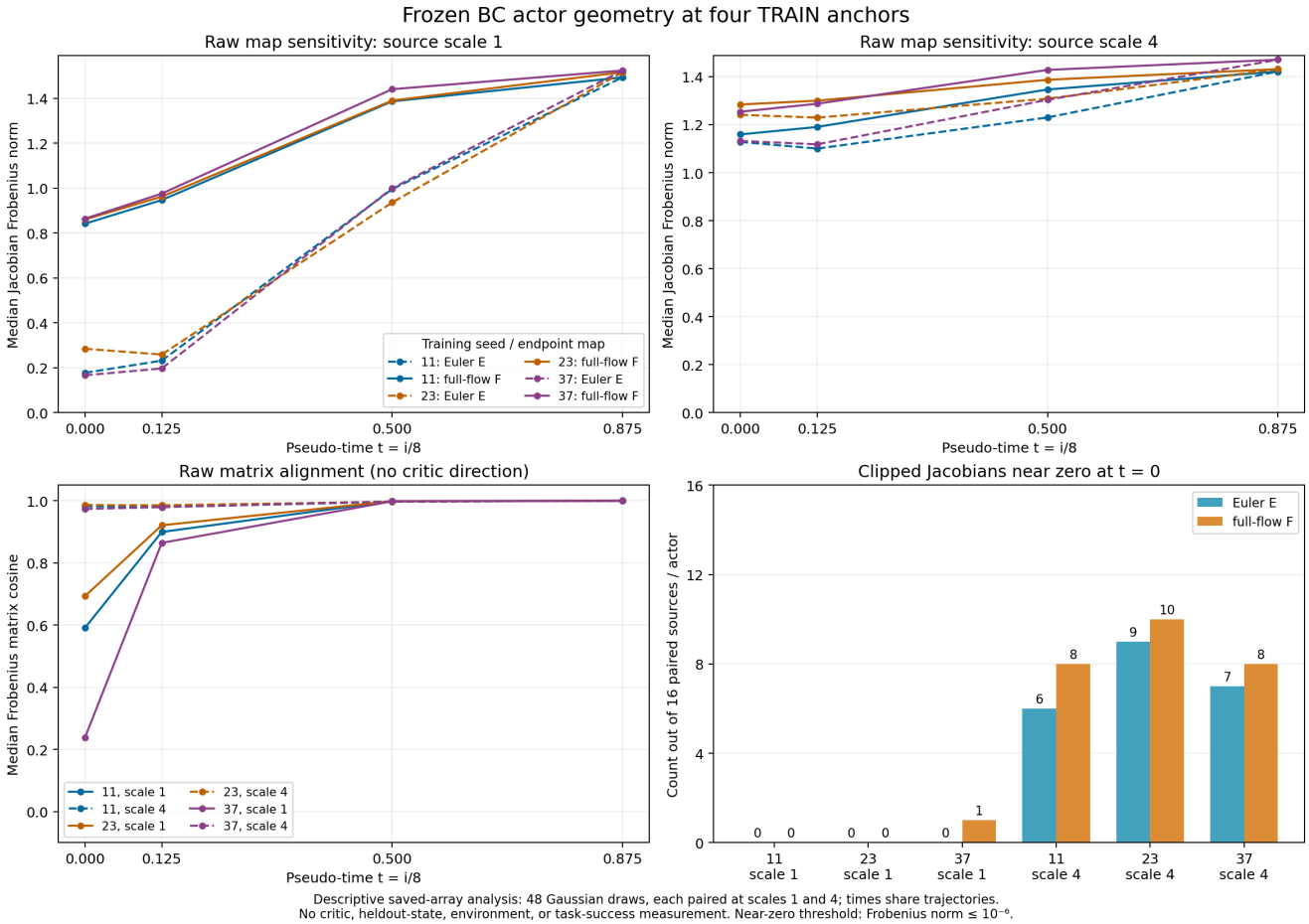
Seed 37 regular Euler L0 primary is BC fallback; its already-frozen positive .08/.1/I8 also 0/8. Small first-action guidance was seen in seed 11 and that secondary, but failed seed 23 Euler actions shifted 1.458/1.223 L2 with predicted raw-Q gains +.400/+.289. Seed 37 failed L1 Euler shifts 1.114/1.446 with gains +.229/+.450. Thus weak first guidance is not the common demonstrated cause. The ideal cancellation identity does not by itself diagnose learned closed-loop failures.

Ordinary IPG remained weak, even where ignored-weight ESS was high. Seed 37 IPG min/median ignored ESS 3.457/15.266 and max solve error .03809 reveal occasional concentration/numerical residual, but earlier failures occurred without that signature. RW shows occasional weight/lineage loss; FK final ESS 16 after resampling conceals pre-resampling minima near 1.3. Raw derivative maxima near 6700 are not centered update magnitudes.

Seed 37 raw/averaged MH have same k80/e.01/I256, 8/8 each, mean acceptance .853/.864 and source displacement 2.479/2.471. Other seeds use different eta/targets; acceptance is not mixing or Q accuracy. Source movement and initial-Q gains are descriptive. Both actor and critic change with trained seed; no causal separation or Q calibration follows. Actor-Jacobian optional status at snapshot: COMPLETED.

# Actor geometry: Euler sensitivity and endpoint clipping

Actor-geometry check (TRAIN inputs only): 48 Gaussian draws from three frozen BC actors were paired at scales 1 and 4. At scale 1 and t=0, median raw Euler/full-flow Jacobian norm ratios were 0.218/0.330/0.205 for seeds 11/23/37. At scale 4 they were 0.968/0.967/0.861, while final clipping made 6/9/7 Euler and 8/10/8 full-flow Jacobians near zero out of 16 per actor. The t=7/8 maps and derivatives agree exactly because one step remains, not because their Jacobians are identity. All 192 predeclared FD rows are retained; no applicable guard failed (maximum normalized error 3.58e-4). This supports a location-dependent geometry mechanism, not a task-failure explanation: no Q direction or selected-loop state was evaluated, and seed 23 had large realized initial shifts in the separate validation diagnosis.




Raw and clipped Jacobians are distinct measurements. At later pseudo-times the derivative is with respect to the CURRENT x_t at an unguided BC prefix, not the original source z. The same 48 draws share their scale and time measurements; 1,536 Jacobian rows are not independent episodes or additional trained seeds.

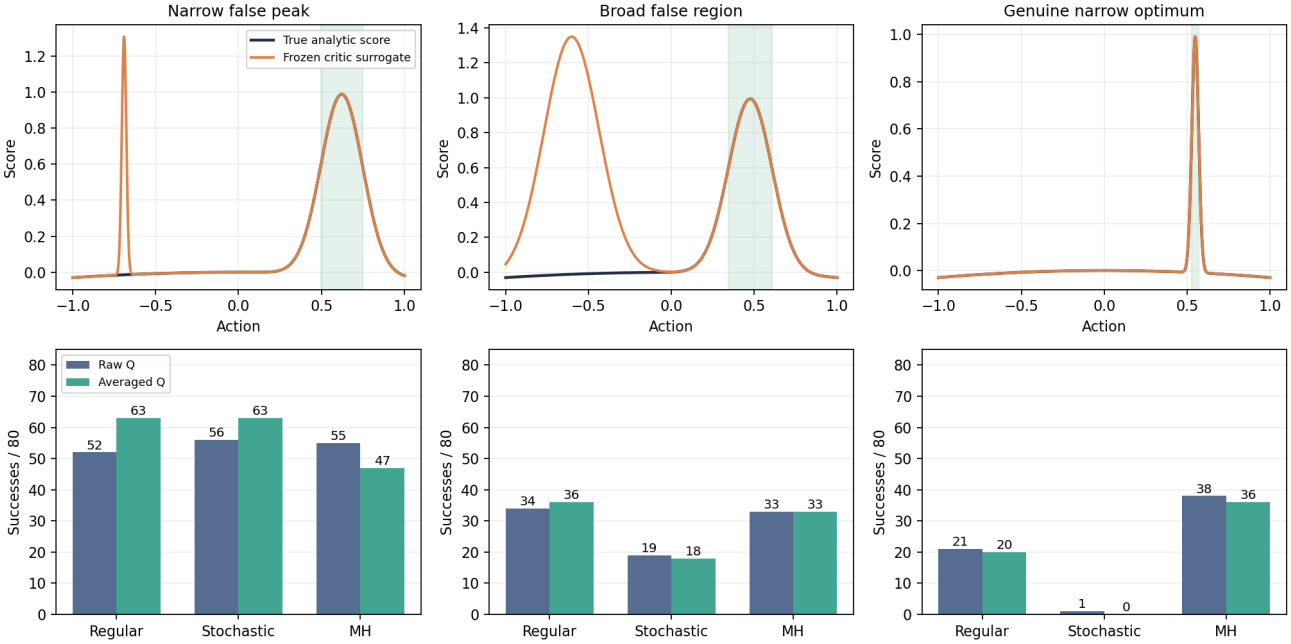
No sampled clip-region crossing or exact boundary occurred in the 192 finite-difference checks, so these passes do not empirically stress the cusp detector. Matrix alignment compares actor Jacobians, not critic-gradient directions. This descriptive TRAIN-input probe supplies neither a causal explanation for heldout failures nor evidence of critic repair.

# Analytic diagnostic: narrow and broad critic errors

Constructed 1D final-point trials use known true scores plus injected critic errors. There are no next states, learned IQL models or physical robot episodes. Twelve cases include narrow false peaks, broad false regions and true narrow optima; validation settings freeze before tests.

| Method | Narrow false /80 | Broad false /80 | True narrow /80 |
|---|---|---|---|
| regular | 52 | 34 | 21 |
| stochastic | 56 | 19 | 1 |
| mh | 55 | 33 | 38 |
| qavg_regular | 63 | 36 | 20 |
| qavg_stochastic | 63 | 18 | 0 |
| qavg_mh | 47 | 33 | 36 |

Controlled score-map diagnostic: averaging helps some cases and hurts others




Synthetic final-point trials, not IQL or robot episodes. Own validation settings; 4 cases × 20 test points per group. Top: first declared case in each group.

Q averaging helped regular/stochastic narrow-false cases, reduced MH there, and did not uniformly repair broad error. It can suppress true narrow optima. Actual unique test points: 1440. These are point outcomes, not trained seeds or physical successes. Own settings confound an isolated smoothing effect.

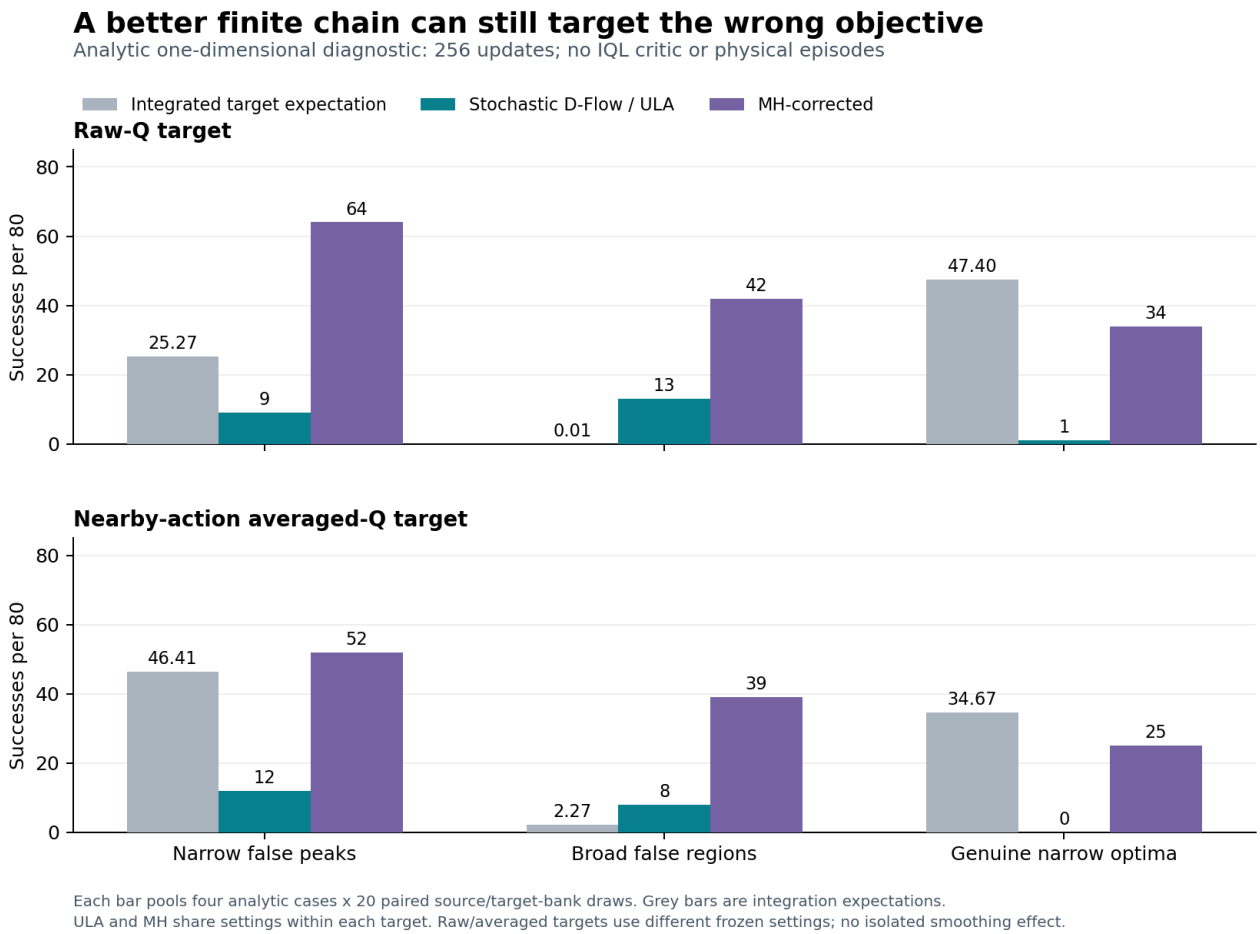
# Matched targets: finite-chain success is not critic repair

| Group/target | Target expected /80 | ULA I256 /80 | MH I256 /80 |
|---|---|---|---|
| narrow false peak/raw | 25.266 | 9 | 64 |
| narrow false peak/qavg | 46.410 | 12 | 52 |
| broad false region/raw | 0.015 | 13 | 42 |
| broad false region/qavg | 2.270 | 8 | 39 |
| genuine narrow optimum/raw | 47.405 | 1 | 34 |
| genuine narrow optimum/qavg | 34.667 | 0 | 25 |

Each target is held fixed from earlier validation selection. Fresh paired sources compare ULA/MH prefixes 1,4,16,64,256, not reselected settings. Broad-false raw target expectation is about .015/80, but finite MH gives 42/80: the finite chain has not equilibrated to the bad critic-defined target. Better finite outcomes can reflect protection from a misleading equilibrium, not repaired Q.

The study has 240 independent case/source random bundles, 960 paired algorithm chains and 4,800 correlated prefix endpoints. It contributes 0 physical episodes and uses 0 learned models. Five endpoints reuse each chain. Integration grid error is 1.07e-05; the tail-probability bound is 2.23e-33.

The true-narrow target expectation falls from 47.405/80 raw to 34.667/80 averaged. Targets differ and have their own frozen parameters, so no isolated causal averaging claim follows. Quadrature-guard failure and missing plotting dependency were preserved; finalization used saved arrays only.




Each bar pools four analytic cases x 20 paired source/target-bank draws. Grey bars are integration expectations.
ULA and MH share settings within each target. Raw/averaged targets use different frozen settings; no isolated smoothing effect.

# Reproducibility, unsuccessful fixes and limits

The exact JSON snapshot binds every used config/cohort/source/checkpoint hash, selection, completed physical metadata/CSV/raw-trace hash and existing audit receipt. It is accompanied by all primary/positive settings, requested test rows, actual counts and file manifest CSVs. The source collector imports no numerical/model/environment library. Raw-array arithmetic and semantic aliases remain separately verified by the independent auditor.

Scientific runtime: CPython 3.11.11/macOS arm64, JAX/JAXlib 0.4.30, Flax 0.8.5, Optax 0.2.4, NumPy 1.26.4, MuJoCo 3.3.7, Gymnasium 0.29.1, OGBench 1.2.1. The inventory provides exact metadata/binary identities and separates the reporting runtime. Inference payload inputs fit about 40 MB before live results. The final manifest specifies whether the original training tables, including the inventoried 49.7 MB data extension, are included; the compiled runtime remains external. Current XML identities are recorded but do not retroactively create launch-time guards.

Preflight caught averaged-Q head-minimum/compiler interaction and batch-shape arithmetic sensitivity. An equivalent two-head elementwise minimum and fixed padded shape passed independent gradients/real-network/compiled checks before physical release. Original eight D-Flow/raw QGF remain unchanged. Numeric int/float identity duplication created redundant executions; they are counted and not presented as independent trials. Failed guards and source revisions remain preserved.

Operational actor/replay watchdog preparation found and fixed Darwin leader/session cleanup edge cases using only short stdlib dummy jobs. Optional actor diagnostic status: COMPLETED; exact replay status: REPLAY_PASS. Prepared does not mean executed. Replay, if completed, repeats the same 20 starts and adds 0 independent tests.

Historical launchers enforce this campaign deadline and completed caches. Full retraining additionally needs the inventoried actual next-state data and original partition/training sources. A small source/results bundle is not a portable offline runtime or guaranteed bitwise reproduction on another platform. Previous campaign/interim reports and frozen archives are preserved; final archive CRC/SHA verification belongs to the closure bundle receipt.

Separate post-deadline reproduction entry-point status: PREPARED_OR_PARTIAL. Preparation/static review does not establish compatibility or a successful replay. Its explicit release and new-output/runtime safeguards remain required. Prior robot manipulation and privileged-demo augmentation results stay in their immutable first-campaign reports and contribute no episodes to this campaign.

Bellman residuals with learned V measure self-consistency, not ground-truth nonterminal Q error. Missing coverage does not prove extrapolation failure; noise, smoothing and MH cannot identify true Q without information. No manipulation competence or general algorithm winner follows from these mazes/analytic points. Unequal search/inference budgets and only 3 trained seeds remain limitations.

Stress-task reset anchors originate from the offline TRAIN table used for learning. Sampler validation/test anchor episodes are disjoint, but the test is not on entirely new learning-state support. The 20 starts are paired across methods and trained seeds; only three training seeds are independent.

The first replay supervisor stopped before creating a child because this Mac's scientific Python 3.11 build lacks os.waitid. A reviewed Anaconda Python 3.13 supervisor override then ran the unchanged Python 3.11 scientific worker. This is an explicit command deviation from the prepared job spec. All 20 trajectories replayed exactly; the future reproduction wrapper remains unexecuted.

Companion compute tables cover completed physical-rollout batch wall/sampler timings only, counted once per actual configuration while retaining numeric rerun costs. Prescreen, training and other work are excluded from those tables; incomplete rollout timing remains unknown. No hardware-normalized Q/NFE query count was measured.

Use a fresh predeclared matched-actor/critic intervention or known-value sequential MDP to separate critic error from proxy-gradient and finite-sampling effects. Keep compute and target definitions matched and use new validation/test cohorts. Current heldout results must not become tuning data.